

The Definite Integrals:

- If $\phi'(x) = f(x)$ then $\int_a^b f(x)dx$ has a definite value $\phi(b) - \phi(a)$, so it is called the definite integral of f from a to b .
- $[a, b]$ is called range of integration while a and b are known as the lower and upper limits respectively.

Properties of Definite Integrals:

(a) $\int_a^b f(x)dx$ gives area bounded by the curve $y = f(x)$ from $x = a$ to $x = b$ and x -axis.

(b) **Fundamental theorem of calculus:** If f is continuous on $[a, b]$ and $\phi'(x) = f(x)$ i.e.

$\phi(x)$ is any anti derivative of f on $[a, b]$, then $\int_a^b f(x)dx = \phi(b) - \phi(a)$

(c) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

(d) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, a < c < b$

(e) The variable of integration x in $\int_a^b f(x)dx$ can be replaced by any other variable

i.e. $\int_a^b f(x)dx = \int_a^b f(t)dt$

Theorem:

(i) **Prove that** $\int_a^b f(x)dx = -\int_b^a f(x)dx$

Proof: If $\phi'(x) = f(x)$ that is if ϕ is an anti derivative of f then using the fundamental theorem of calculus, we get

$$\begin{aligned}\int_a^b f(x)dx &= \phi(b) - \phi(a) \\ &= -[\phi(a) - \phi(b)] \\ \int_a^b f(x)dx &= -\int_b^a f(x)dx\end{aligned}$$

(ii) Prove that $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$; $a < c < b$

Proof: If $\phi'(x) = f(x)$ that is if $\phi(x)$ is an anti derivative of $f(x)$, then applying the fundamental theorem of calculus we have

$$\int_a^c f(x)dx = \phi(c) - \phi(a) \dots\dots(i)$$

and

$$\int_c^b f(x)dx = \phi(b) - \phi(c) \dots\dots(ii)$$

Adding (i) and (ii)

$$\begin{aligned} \int_a^c f(x)dx + \int_c^b f(x)dx &= \phi(c) - \phi(a) + \phi(b) - \phi(c) \\ &= \phi(b) - \phi(a) \\ &= \int_a^b f(x)dx. \end{aligned}$$

EXERCISE 3.6

Evaluate the following definite integrals:

Q.1 $\int_1^2 (x^2 + 1)dx$

Solution:

$$\begin{aligned} &\int_1^2 (x^2 + 1)dx \\ &= \int_1^2 x^2 dx + \int_1^2 1 dx \\ &= \left[\frac{x^3}{3} \right]_1^2 + [x]_1^2 \\ &= \frac{1}{3} [x^3]_1^2 + [x]_1^2 \\ &= \frac{1}{3} [2^3 - 1^3] + [2 - 1] \\ &= \frac{1}{3} [8 - 1] + 1 \\ &= \frac{7}{3} + 1 \\ &= \frac{7+3}{3} \\ &= \frac{10}{3} \end{aligned}$$

Q.2 $\int_{-1}^1 \left(x^{\frac{1}{3}} + 1 \right) dx$

Solution:

$$\begin{aligned} &\int_{-1}^1 \left(x^{\frac{1}{3}} + 1 \right) dx \\ &= \int_{-1}^1 x^{\frac{1}{3}} dx + \int_{-1}^1 1 dx \\ &= \left[\frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} \right]_{-1}^1 + [x]_{-1}^1 \\ &= \left[\frac{x^{\frac{4}{3}}}{\frac{4}{3}} \right]_{-1}^1 + [1 - (-1)] \\ &= \frac{3}{4} \left[x^{\frac{4}{3}} \right]_{-1}^1 + [1 + 1] \\ &= \frac{3}{4} \left[1^{\frac{4}{3}} - (-1)^{\frac{4}{3}} \right] + 2 \end{aligned}$$

$$= \frac{3}{4} \left[1 - \left\{ (-1)^4 \right\}^{\frac{1}{3}} \right] + 2$$

$$= \frac{3}{4} \left[1 - 1^{\frac{1}{3}} \right] + 2$$

$$= \frac{3}{4} [1 - 1] + 2$$

$$= \frac{3}{4} (0) + 2$$

$$= 0 + 2$$

$$= 2$$

Q.3 $\int_{-2}^0 \frac{1}{(2x-1)^2} dx$

Solution:

$$\int_{-2}^0 \frac{1}{(2x-1)^2} dx$$

$$= \int_{-2}^0 (2x-1)^{-2} dx$$

$$= \frac{1}{2} \int_{-2}^0 (2x-1)^{-2} \cdot 2 dx$$

$$= \frac{1}{2} \left[\frac{(2x-1)^{-2+1}}{-2+1} \right]_{-2}^0$$

$$= \frac{1}{2} \left[\frac{(2x-1)^{-1}}{-1} \right]_{-2}^0$$

$$= -\frac{1}{2} \left[\frac{1}{2x-1} \right]_{-2}^0$$

$$= -\frac{1}{2} \left[\frac{1}{2(0)-1} - \frac{1}{2(-2)-1} \right]$$

$$= -\frac{1}{2} \left[\frac{1}{-1} - \frac{1}{-5} \right]$$

$$= -\frac{1}{2} \left[-1 + \frac{1}{5} \right]$$

$$= -\frac{1}{2} \left[\frac{-5+1}{5} \right]$$

$$= -\frac{1}{2} \left[\frac{-4}{5} \right]$$

$$= \frac{2}{5}$$

Q.4 $\int_{-6}^2 \sqrt{3-x} dx$

Solution:

$$\int_{-6}^2 \sqrt{3-x} dx$$

$$= (-1) \int_{-6}^2 \sqrt{3-x} (-1) dx$$

$$= (-1) \left[\frac{(3-x)^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_{-6}^2$$

$$= (-1) \left[\frac{(3-x)^{\frac{3}{2}}}{\frac{3}{2}} \right]_{-6}^2$$

$$= -\frac{2}{3} \left[(3-2)^{\frac{3}{2}} - (3-(-6))^{\frac{3}{2}} \right]$$

$$= -\frac{2}{3} \left[1^{\frac{3}{2}} - 9^{\frac{3}{2}} \right]$$

$$= -\frac{2}{3} \left[1 - (3^2)^{\frac{3}{2}} \right]$$

$$= -\frac{2}{3} [1 - 3^3]$$

$$= -\frac{2}{3} [1 - 27]$$

$$= -\frac{2}{3} [-26]$$

$$= \frac{52}{3}$$

Q.5 $\int_1^{\sqrt{5}} \sqrt{(2t-1)^3} dt$

Solution:

$$\int_1^{\sqrt{5}} \sqrt{(2t-1)^3} dt$$

$$= \frac{1}{2} \int_1^{\sqrt{5}} (2t-1)^{\frac{3}{2}} \cdot 2 dt$$

$$\begin{aligned}
&= \frac{1}{2} \left[\frac{(2t-1)^{\frac{3}{2}+1}}{\frac{3}{2}+1} \right]_1^{\sqrt{5}} \\
&= \frac{1}{2} \left[\frac{(2t-1)^{\frac{5}{2}}}{\frac{5}{2}} \right]_1^{\sqrt{5}} \\
&= \frac{1}{5} \left[(2t-1)^{\frac{5}{2}} \right]_1^{\sqrt{5}} \\
&= \frac{1}{5} \left[(2\sqrt{5}-1)^{\frac{5}{2}} - (2(1)-1)^{\frac{5}{2}} \right] \\
&= \frac{1}{5} \left[(2\sqrt{5}-1)^{\frac{5}{2}} - 1 \right]
\end{aligned}$$

Q.6 $\int_2^{\sqrt{5}} x\sqrt{x^2-1} dx$

Solution:

$$\begin{aligned}
&\int_2^{\sqrt{5}} x\sqrt{x^2-1} dx \\
&= \frac{1}{2} \int_2^{\sqrt{5}} \sqrt{x^2-1} \cdot (2x) dx \\
&= \frac{1}{2} \left[\frac{(x^2-1)^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_2^{\sqrt{5}} \\
&= \frac{1}{2} \left[\frac{(x^2-1)^{\frac{3}{2}}}{\frac{3}{2}} \right]_2^{\sqrt{5}} \\
&= \frac{1}{3} \left[(x^2-1)^{\frac{3}{2}} \right]_2^{\sqrt{5}} \\
&= \frac{1}{3} \left[((\sqrt{5})^2-1)^{\frac{3}{2}} - (2^2-1)^{\frac{3}{2}} \right] \\
&= \frac{1}{3} \left[(5-1)^{\frac{3}{2}} - (4-1)^{\frac{3}{2}} \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{3} \left[4^{\frac{3}{2}} - 3^{\frac{3}{2}} \right] \\
&= \frac{1}{3} \left[(2^2)^{\frac{3}{2}} - 3 \cdot 3^{\frac{1}{2}} \right] \\
&= \frac{1}{3} [2^3 - 3\sqrt{3}] \\
&= \frac{1}{3} [8 - 3\sqrt{3}]
\end{aligned}$$

Q.7 $\int_1^2 \frac{x}{x^2+2} dx$

Solution:

$$\begin{aligned}
&\int_1^2 \frac{x}{x^2+2} dx \\
&= \frac{1}{2} \int_1^2 \frac{2x}{x^2+2} dx \\
&= \frac{1}{2} \left[\ln(x^2+2) \right]_1^2 \\
&= \frac{1}{2} [\ln(2^2+2) - \ln(1^2+2)] \\
&= \frac{1}{2} [\ln 6 - \ln 3] \\
&= \frac{1}{2} \left[\ln \left(\frac{6}{3} \right) \right] \\
&= \frac{1}{2} \ln 2
\end{aligned}$$

Q.8 $\int_2^3 \left(x - \frac{1}{x} \right)^2 dx$

Solution:

$$\begin{aligned}
&\int_2^3 \left(x - \frac{1}{x} \right)^2 dx \\
&= \int_2^3 \left(x^2 + \frac{1}{x^2} - 2x \frac{1}{x} \right) dx \\
&= \int_2^3 (x^2 + x^{-2} - 2) dx \\
&= \int_2^3 x^2 dx + \int_2^3 x^{-2} dx - 2 \int_2^3 1 dx
\end{aligned}$$

$$\begin{aligned}
&= \left[\frac{x^{2+1}}{2+1} \right]_2^3 + \left[\frac{x^{-2+1}}{-2+1} \right]_2^3 - 2[x]_2^3 \\
&= \left[\frac{x^3}{3} \right]_2^3 + \left[\frac{x^{-1}}{-1} \right]_2^3 - 2[3-2] \\
&= \frac{1}{3} \left[x^3 \right]_2^3 - \left[\frac{1}{x} \right]_2^3 - 2[1] \\
&= \frac{1}{3} [3^3 - 2^3] - \left[\frac{1}{3} - \frac{1}{2} \right] - 2 \\
&= \frac{1}{3} [27 - 8] - \left[\frac{2-3}{6} \right] - 2 \\
&= \frac{19}{3} - \left(\frac{-1}{6} \right) - 2 \\
&= \frac{19}{3} + \frac{1}{6} - 2 \\
&= \frac{38+1-12}{6} \\
&= \frac{27}{6} \\
&= \frac{9}{2}
\end{aligned}$$

Q.9 $\int_{-1}^1 \left(x + \frac{1}{2} \right) \sqrt{x^2 + x + 1} dx$

Solution:

$$\begin{aligned}
&\int_{-1}^1 \left(x + \frac{1}{2} \right) \sqrt{x^2 + x + 1} dx \\
&= \int_{-1}^1 \left(\frac{2x+1}{2} \right) \sqrt{x^2 + x + 1} dx \\
&= \frac{1}{2} \int_{-1}^1 (x^2 + x + 1)^{\frac{1}{2}} (2x+1) dx \\
&= \frac{1}{2} \left[\frac{(x^2 + x + 1)^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_{-1}^1 \\
&= \frac{1}{2} \left[\frac{(x^2 + x + 1)^{\frac{3}{2}}}{\frac{3}{2}} \right]_{-1}^1
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{3} \left[(x^2 + x + 1)^{\frac{3}{2}} \right]_{-1}^1 \\
&= \frac{1}{3} \left[(1^2 + 1 + 1)^{\frac{3}{2}} - ((-1)^2 + (-1) + 1)^{\frac{3}{2}} \right] \\
&= \frac{1}{3} \left[3^{\frac{3}{2}} - (1 - 1 + 1)^{\frac{3}{2}} \right] \\
&= \frac{1}{3} \left[3 \cdot 3^{\frac{1}{2}} - 1^{\frac{3}{2}} \right] \\
&= \frac{1}{3} [3\sqrt{3} - 1] \\
&= \sqrt{3} - \frac{1}{3}
\end{aligned}$$

Q.10 $\int_0^3 \frac{dx}{x^2 + 9}$

Solution:

$$\begin{aligned}
&\int_0^3 \frac{1}{x^2 + 9} dx \\
&= \int_0^3 \frac{1}{x^2 + 3^2} dx \\
&= \frac{1}{3} \left[\tan^{-1} \left(\frac{x}{3} \right) \right]_0^3 \\
&= \frac{1}{3} \left[\tan^{-1} \left(\frac{3}{3} \right) - \tan^{-1} \left(\frac{0}{3} \right) \right] \\
&= \frac{1}{3} [\tan^{-1}(1) - \tan^{-1}(0)] \\
&= \frac{1}{3} \left[\frac{\pi}{4} - 0 \right] \\
&= \frac{\pi}{12}
\end{aligned}$$

Q.11 $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos t \, dt$

Solution:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos t \, dt$$

$$\begin{aligned}
 &= \left[\sin t \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\
 &= \sin \frac{\pi}{3} - \sin \frac{\pi}{6} \\
 &= \frac{\sqrt{3}}{2} - \frac{1}{2} \\
 &= \frac{\sqrt{3}-1}{2}
 \end{aligned}$$

Q.12 $\int_1^2 \left(x + \frac{1}{x} \right)^{\frac{1}{2}} \left(1 - \frac{1}{x^2} \right) dx$

Solution:

$$\begin{aligned}
 &\int_1^2 \left(x + \frac{1}{x} \right)^{\frac{1}{2}} \left(1 - \frac{1}{x^2} \right) dx \\
 &= \left[\frac{\left(x + \frac{1}{x} \right)^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_1^2 \\
 &= \left[\frac{\left(x + \frac{1}{x} \right)^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^2 \\
 &= \frac{2}{3} \left[\left(x + \frac{1}{x} \right)^{\frac{3}{2}} \right]_1^2 \\
 &= \frac{2}{3} \left[\left(2 + \frac{1}{2} \right)^{\frac{3}{2}} - \left(1 + \frac{1}{1} \right)^{\frac{3}{2}} \right] \\
 &= \frac{2}{3} \left[\left(\frac{4+1}{2} \right)^{\frac{3}{2}} - 2^{\frac{3}{2}} \right] \\
 &= \frac{2}{3} \left[\left(\frac{5}{2} \right)^{\frac{3}{2}} - 2^{\frac{3}{2}} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{3} \left[\frac{5^{\frac{3}{2}}}{2^{\frac{3}{2}}} - 2^{\frac{3}{2}} \right] \\
 &= \frac{2}{3} \left[\frac{5^{\frac{3}{2}} - 2^3}{2^{\frac{3}{2}}} \right] \\
 &= \frac{2}{3} \left[\frac{5 \cdot 5^{\frac{1}{2}} - 8}{2 \cdot 2^{\frac{1}{2}}} \right] \\
 &= \frac{1}{3\sqrt{2}} [5\sqrt{5} - 8]
 \end{aligned}$$

Q.13 $\int_1^2 \ln x \, dx$

Solution:

$$\begin{aligned}
 &\int_1^2 \ln x \, dx \\
 &= \int_1^2 \ln x \cdot 1 \, dx \\
 &\text{Integrating by parts} \\
 &= [\ln x \cdot x]_1^2 - \int_1^2 x \cdot \frac{1}{x} \, dx \\
 &= 2 \ln 2 - 1 \cdot \ln(1) - \int_1^2 1 \, dx \\
 &= 2 \ln 2 - 0 - [x]_1^2 \\
 &= 2 \ln 2 - (2 - 1) \\
 &= 2 \ln 2 - 1
 \end{aligned}$$

Q.14 $\int_0^2 \left(e^{\frac{x}{2}} - e^{\frac{-x}{2}} \right) dx$

Solution:

$$\begin{aligned}
 &\int_0^2 \left(e^{\frac{x}{2}} - e^{\frac{-x}{2}} \right) dx \\
 &= \int_0^2 e^{\frac{x}{2}} \, dx - \int_0^2 e^{\frac{-x}{2}} \, dx \\
 &= 2 \left[\frac{e^{\frac{x}{2}}}{\frac{1}{2}} \right]_0^2 - 2 \left[\frac{e^{\frac{-x}{2}}}{\frac{-1}{2}} \right]_0^2 \\
 &= 2 \left[\frac{e^{\frac{x}{2}}}{\frac{1}{2}} \right]_0^2 - 2 \left[\frac{e^{\frac{-x}{2}}}{\frac{-1}{2}} \right]_0^2
 \end{aligned}$$

$$\begin{aligned}
&= 2 \left[e^{\frac{x}{2}} \right]_0^2 + 2 \left[e^{\frac{-x}{2}} \right]_0^2 \\
&= 2 \left[e^{\frac{2}{2}} - e^{\frac{0}{2}} \right] + 2 \left[e^{\frac{-2}{2}} - e^{\frac{0}{2}} \right] \\
&= 2 \left[e^1 - e^0 \right] + 2 \left[e^{-1} - e^0 \right] \\
&= 2 \left[e - 1 \right] + 2 \left[\frac{1}{e} - 1 \right] \\
&= 2 \left[e - 1 + \frac{1}{e} - 1 \right] \\
&= 2 \left[e + \frac{1}{e} - 2 \right] \\
&= 2 \left[\frac{e^2 + 1 - 2e}{e} \right] \\
&= 2 \left[\frac{(e-1)^2}{e} \right] \\
&= \frac{2}{e} (e-1)^2
\end{aligned}$$

Q.15 $\int_0^{\frac{\pi}{4}} \frac{\cos \theta + \sin \theta}{2 \cos^2 \theta} d\theta$

Solution:

$$\begin{aligned}
&\int_0^{\frac{\pi}{4}} \frac{\cos \theta + \sin \theta}{2 \cos^2 \theta} d\theta \\
&= \int_0^{\frac{\pi}{4}} \left[\frac{\cos \theta}{2 \cos^2 \theta} + \frac{\sin \theta}{2 \cos^2 \theta} \right] d\theta \\
&= \int_0^{\frac{\pi}{4}} \left[\frac{1}{2 \cos \theta} + \frac{\sin \theta}{2 \cos \theta \cos \theta} \right] d\theta \\
&= \frac{1}{2} \int_0^{\frac{\pi}{4}} \sec \theta d\theta + \frac{1}{2} \int_0^{\frac{\pi}{4}} \sec \theta \tan \theta d\theta \\
&= \frac{1}{2} \left[\ln |\sec \theta + \tan \theta| \right]_0^{\frac{\pi}{4}} + \frac{1}{2} \left[\sec \theta \right]_0^{\frac{\pi}{4}} \\
&= \frac{1}{2} \left[\ln \left| \sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right| - \ln |\sec 0 + \tan 0| \right] + \frac{1}{2} \left[\sec \frac{\pi}{4} - \sec 0 \right] \\
&= \frac{1}{2} \left[\ln |\sqrt{2} + 1| - \ln |1 + 0| \right] + \frac{1}{2} \left[\sqrt{2} - 1 \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \left[\ln |\sqrt{2} + 1| - 0 \right] + \frac{1}{2} \left[\sqrt{2} - 1 \right] \\
&= \frac{1}{2} \ln |\sqrt{2} + 1| + \frac{1}{2} \left[\sqrt{2} - 1 \right] \\
&= \frac{1}{2} \left[\ln (\sqrt{2} + 1) + \sqrt{2} - 1 \right]
\end{aligned}$$

Q.16 $\int_0^{\frac{\pi}{6}} \cos^3 \theta d\theta$

Solution:

$$\begin{aligned}
&\int_0^{\frac{\pi}{6}} \cos^3 \theta d\theta \\
&= \int_0^{\frac{\pi}{6}} \cos^2 \theta \cdot \cos \theta d\theta \\
&= \int_0^{\frac{\pi}{6}} (1 - \sin^2 \theta) \cos \theta d\theta \\
&= \int_0^{\frac{\pi}{6}} (\cos \theta - \sin^2 \theta \cos \theta) d\theta \\
&= \int_0^{\frac{\pi}{6}} \cos \theta d\theta - \int_0^{\frac{\pi}{6}} \sin^2 \theta \cos \theta d\theta \\
&= \left[\sin \theta \right]_0^{\frac{\pi}{6}} - \left[\frac{\sin^3 \theta}{3} \right]_0^{\frac{\pi}{6}} \\
&= \left[\sin \frac{\pi}{6} - \sin 0 \right] - \frac{1}{3} \left[\sin^3 \frac{\pi}{6} - \sin^3 0 \right] \\
&= \left[\frac{1}{2} - 0 \right] - \frac{1}{3} \left[\left(\frac{1}{2} \right)^3 - 0 \right] \\
&= \frac{1}{2} - \frac{1}{3} \left(\frac{1}{8} \right) \\
&= \frac{1}{2} - \frac{1}{24} \\
&= \frac{12-1}{24} \\
&= \frac{11}{24}
\end{aligned}$$

Q.17 $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos^2 \theta \cot^2 \theta d\theta$

Solution:

$$\begin{aligned}
 & \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos^2 \theta \cot^2 \theta d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos^2 \theta (\operatorname{cosec}^2 \theta - 1) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos^2 \theta \left(\frac{1}{\sin^2 \theta} - 1 \right) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(\frac{\cos^2 \theta}{\sin^2 \theta} - \cos^2 \theta \right) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\cot^2 \theta - \cos^2 \theta) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cot^2 \theta d\theta - \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos^2 \theta d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\operatorname{cosec}^2 \theta - 1) d\theta - \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \operatorname{cosec}^2 \theta d\theta - \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 1 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 1 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \cos 2\theta d\theta \\
 &= \left[-\cot \theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} - \left[\theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} - \frac{1}{2} \left[\theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} \\
 &= - \left[\cot \frac{\pi}{4} - \cot \frac{\pi}{6} \right] - \left[\frac{\pi}{4} - \frac{\pi}{6} \right] - \frac{1}{2} \left[\frac{\pi}{4} - \frac{\pi}{6} \right] \\
 &\quad - \frac{1}{4} \left[\sin \left(\frac{2\pi}{4} \right) - \sin \left(\frac{2\pi}{6} \right) \right]
 \end{aligned}$$

$$\begin{aligned}
 &= - \left[1 - \sqrt{3} \right] - \left[\frac{3\pi - 2\pi}{12} \right] - \frac{1}{2} \left[\frac{3\pi - 2\pi}{12} \right] \\
 &\quad - \frac{1}{4} \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{3} \right] \\
 &= -1 + \sqrt{3} - \frac{\pi}{12} - \frac{1}{2} \left(\frac{\pi}{12} \right) - \frac{1}{4} \left[1 - \frac{\sqrt{3}}{2} \right] \\
 &= -1 + \sqrt{3} - \frac{\pi}{12} - \frac{\pi}{24} - \frac{1}{4} + \frac{\sqrt{3}}{8} \\
 &= -1 + \sqrt{3} - \frac{1}{4} + \frac{\sqrt{3}}{8} - \left[\frac{\pi}{12} + \frac{\pi}{24} \right] \\
 &= \frac{-8 + 8\sqrt{3} - 2 + \sqrt{3}}{8} - \left[\frac{2\pi + \pi}{24} \right] \\
 &= \frac{-10 + 9\sqrt{3}}{8} - \frac{3\pi}{24} \\
 &= \frac{-10 + 9\sqrt{3}}{8} - \frac{\pi}{8} \\
 &= \frac{9\sqrt{3} - 10 - \pi}{8}
 \end{aligned}$$

Q.18 $\int_0^{\frac{\pi}{4}} \cos^4 t dt$

Solution:

$$\begin{aligned}
 & \int_0^{\frac{\pi}{4}} (\cos^2 t)^2 dt \\
 &= \int_0^{\frac{\pi}{4}} \left(\frac{1 + \cos 2t}{2} \right)^2 dt \\
 &= \frac{1}{4} \int_0^{\frac{\pi}{4}} [1 + \cos^2 2t + 2 \cos 2t] dt \\
 &= \frac{1}{4} \int_0^{\frac{\pi}{4}} 1 dt + \frac{1}{4} \int_0^{\frac{\pi}{4}} \cos^2 2t dt + \frac{2}{4} \int_0^{\frac{\pi}{4}} \cos 2t dt \\
 &= \frac{1}{4} [t]_0^{\frac{\pi}{4}} + \frac{1}{4} \int_0^{\frac{\pi}{4}} \frac{1 + \cos 4t}{2} dt + \frac{1}{2} \left[\frac{\sin 2t}{2} \right]_0^{\frac{\pi}{4}}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \left[\frac{\pi}{4} - 0 \right] + \frac{1}{8} \int_0^{\frac{\pi}{4}} (1 + \cos 4t) dt + \frac{1}{4} [\sin 2t]_0^{\frac{\pi}{4}} \\
&= \frac{\pi}{16} + \frac{1}{8} \left[t + \frac{\sin 4t}{4} \right]_0^{\frac{\pi}{4}} + \frac{1}{4} \left[\sin 2 \left(\frac{\pi}{4} \right) - \sin 2(0) \right] \\
&= \frac{\pi}{16} + \frac{1}{8} \left[\frac{\pi}{4} + \frac{\sin 4 \left(\frac{\pi}{4} \right)}{4} \right] - \frac{1}{8} \left[0 + \frac{\sin 4(0)}{4} \right] + \frac{1}{4} \left[\sin \frac{\pi}{2} - 0 \right] \\
&= \frac{\pi}{16} + \frac{1}{8} \left[\frac{\pi}{4} + 0 \right] - \frac{1}{8} [0 + 0] + \frac{1}{4} [1 - 0] \\
&= \frac{\pi}{16} + \frac{\pi}{32} + \frac{1}{4} \\
&= \frac{2\pi + \pi + 8}{32} = \frac{3\pi + 8}{32}
\end{aligned}$$

Q.19 $\int_0^{\frac{\pi}{3}} \cos^2 \theta \sin \theta d\theta$

Solution:

$$\begin{aligned}
&\int_0^{\frac{\pi}{3}} \cos^2 \theta \sin \theta d\theta \\
&= - \int_0^{\frac{\pi}{3}} \cos^2 \theta (-\sin \theta) d\theta \\
&= - \left[\frac{\cos^3 \theta}{3} \right]_0^{\frac{\pi}{3}} \\
&= - \frac{1}{3} \left[\cos^3 \theta \right]_0^{\frac{\pi}{3}} \\
&= - \frac{1}{3} \left[\cos^3 \frac{\pi}{3} - \cos^3 0 \right] \\
&= - \frac{1}{3} \left[\left(\frac{1}{2} \right)^3 - 1^3 \right] \\
&= - \frac{1}{3} \left[\frac{1}{8} - 1 \right] \\
&= - \frac{1}{3} \left[\frac{1-8}{8} \right] \\
&= - \frac{1}{3} \left[\frac{-7}{8} \right] \\
&= \frac{7}{24}
\end{aligned}$$

Q.20 $\int_0^{\frac{\pi}{4}} (1 + \cos^2 \theta) \tan^2 \theta d\theta$

Solution:

$$\begin{aligned}
&\int_0^{\frac{\pi}{4}} (1 + \cos^2 \theta) \tan^2 \theta d\theta \\
&= \int_0^{\frac{\pi}{4}} (\tan^2 \theta + \cos^2 \theta \tan^2 \theta) d\theta \\
&= \int_0^{\frac{\pi}{4}} (\tan^2 \theta + \sin^2 \theta) d\theta \\
&= \int_0^{\frac{\pi}{4}} \tan^2 \theta d\theta + \int_0^{\frac{\pi}{4}} \sin^2 \theta d\theta \\
&= \int_0^{\frac{\pi}{4}} (\sec^2 \theta - 1) d\theta + \int_0^{\frac{\pi}{4}} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \\
&= \int_0^{\frac{\pi}{4}} \sec^2 \theta d\theta - \int_0^{\frac{\pi}{4}} 1 d\theta + \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 d\theta - \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2\theta d\theta \\
&= [\tan \theta]_0^{\frac{\pi}{4}} - [\theta]_0^{\frac{\pi}{4}} + \frac{1}{2} [\theta]_0^{\frac{\pi}{4}} - \frac{1}{2} \left[\frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}} \\
&= \left[\tan \frac{\pi}{4} - \tan 0 \right] - \left[\frac{\pi}{4} - 0 \right] + \frac{1}{2} \left[\frac{\pi}{4} - 0 \right] \\
&\quad - \frac{1}{4} \left[\sin 2 \left(\frac{\pi}{4} \right) - \sin 2(0) \right] \\
&= [1 - 0] - \frac{\pi}{4} + \frac{\pi}{8} - \frac{1}{4} \left[\sin \frac{\pi}{2} - \sin 0 \right] \\
&= 1 - \frac{\pi}{4} + \frac{\pi}{8} - \frac{1}{4} [1 - 0] \\
&= 1 - \frac{\pi}{4} + \frac{\pi}{8} - \frac{1}{4} \\
&= \frac{8 - 2\pi + \pi - 2}{8} \\
&= \frac{6 - \pi}{8}
\end{aligned}$$

Q.21 $\int_0^{\frac{\pi}{4}} \frac{\sec \theta}{\sin \theta + \cos \theta} d\theta$

Solution:

$$\begin{aligned} & \int_0^{\frac{\pi}{4}} \frac{\sec \theta}{\sin \theta + \cos \theta} d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{\sec \theta d\theta}{\cos \theta \left(\frac{\sin \theta}{\cos \theta} + 1 \right)} \\ &= \int_0^{\frac{\pi}{4}} \frac{\sec \theta \cdot \sec \theta d\theta}{\tan \theta + 1} \\ &= \int_0^{\frac{\pi}{4}} \frac{\sec^2 \theta d\theta}{\tan \theta + 1} \end{aligned}$$

Using $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$

$$\begin{aligned} &= \left[\ln(\tan \theta + 1) \right]_0^{\frac{\pi}{4}} \\ &= \ln \left(\tan \frac{\pi}{4} + 1 \right) - \ln(\tan 0 + 1) \\ &= \ln(1+1) - \ln(0+1) \\ &= \ln 2 - \ln 1 \\ &= \ln 2 \quad \because \ln 1 = 0 \end{aligned}$$

Q.22 $\int_{-1}^5 |x-3| dx$

Solution:

$$\int_{-1}^5 |x-3| dx$$

Using $|x-3| = \begin{cases} x-3 & \text{if } x \geq 3 \\ -(x-3) & \text{if } x < 3 \end{cases}$

$$\begin{aligned} &= -\int_{-1}^3 (x-3) dx + \int_3^5 (x-3) dx \\ &= -\left[\frac{x^2}{2} - 3x \right]_{-1}^3 + \left[\frac{x^2}{2} - 3x \right]_3^5 \end{aligned}$$

$$\begin{aligned} &= -\left\{ \left(\frac{9}{2} - 9 \right) - \left(\frac{1}{2} + 3 \right) \right\} + \left\{ \left(\frac{25}{2} - 15 \right) - \left(\frac{9}{2} - 9 \right) \right\} \\ &= -\left\{ \frac{-9}{2} - \frac{7}{2} \right\} + \left\{ \frac{-5}{2} + \frac{9}{2} \right\} \\ &= -\left\{ \frac{-16}{2} \right\} + \left\{ \frac{4}{2} \right\} \\ &= 8 + 2 \\ &= 10 \end{aligned}$$

Q.23 $\int_{\frac{1}{8}}^1 \frac{\left(x^{\frac{1}{3}} + 2\right)^2}{x^{\frac{2}{3}}} dx$

Solution:

$$\begin{aligned} & \int_{\frac{1}{8}}^1 \frac{\left(x^{\frac{1}{3}} + 2\right)^2}{x^{\frac{2}{3}}} dx \\ &= 3 \int_{\frac{1}{8}}^1 \left(x^{\frac{1}{3}} + 2\right)^2 \left(\frac{1}{3} x^{-\frac{2}{3}}\right) dx \end{aligned}$$

Using $\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + c$

$$\begin{aligned} &= 3 \left[\frac{\left(x^{\frac{1}{3}} + 2\right)^3}{3} \right]_{\frac{1}{8}}^1 \\ &= \frac{3}{3} \left[\left\{ \left(1\right)^{\frac{1}{3}} + 2 \right\}^3 - \left\{ \left(\frac{1}{8}\right)^{\frac{1}{3}} + 2 \right\}^3 \right] \\ &= (1+2)^3 - \left(\frac{1}{2} + 2\right)^3 \\ &= 27 - \frac{125}{8} \\ &= \frac{216-125}{8} \\ &= \frac{91}{8} \end{aligned}$$

Q.24 $\int_1^3 \frac{x^2-2}{x+1} dx$

Solution:

$$\int_1^3 \frac{x^2-2}{x+1} dx$$

By long division

$$\begin{array}{r} x-1 \\ x+1 \overline{) x^2-2} \\ \underline{+x^2+x} \\ -x-2 \\ \underline{+x+1} \\ -1 \end{array}$$

$$= \int_1^3 \left(x-1 - \frac{1}{x+1} \right) dx$$

$$= \int_1^3 x dx - \int_1^3 1 dx - \int_1^3 \frac{1}{x+1} dx$$

$$= \left[\frac{x^2}{2} \right]_1^3 - [x]_1^3 - [\ln(x+1)]_1^3$$

$$= \left[\frac{9}{2} - \frac{1}{2} \right] - (3-1) - [\ln(3+1) - \ln(1+1)]$$

$$= 4 - 2 - \ln 4 + \ln 2$$

$$= 2 - \ln\left(\frac{4}{2}\right) = 2 - \ln(2)$$

Q.25 $\int_2^3 \frac{3x^2-2x+1}{(x-1)(x^2+1)} dx$

Solution:

$$\int_2^3 \frac{3x^2-2x+1}{(x-1)(x^2+1)} dx$$

$$= \int_2^3 \frac{3x^2-2x+1}{x^3-x^2+x-1} dx$$

$$= \int_2^3 \frac{d}{dx} \frac{(x^3-x^2+x-1)}{x^3-x^2+x-1} dx$$

$$= [\ln(x^3-x^2+x-1)]_2^3$$

$$= \ln(3^3-3^2+3-1) - \ln(2^3-2^2+2-1)$$

$$= \ln(27-9+2) - \ln(8-4+1)$$

$$= \ln 20 - \ln 4$$

$$= \ln \frac{20}{4} = \ln 5$$

Q.26 $\int_0^{\frac{\pi}{4}} \frac{\sin x - 1}{\cos^2 x} dx$

Solution:

$$\int_0^{\frac{\pi}{4}} \frac{\sin x - 1}{\cos^2 x} dx$$

$$= \int_0^{\frac{\pi}{4}} \left(\frac{\sin x}{\cos^2 x} - \frac{1}{\cos^2 x} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \sec x \tan x dx - \int_0^{\frac{\pi}{4}} \sec^2 x dx$$

$$= [\sec x]_0^{\frac{\pi}{4}} - [\tan x]_0^{\frac{\pi}{4}}$$

$$= \sec\left(\frac{\pi}{4}\right) - \sec(0) - \left(\tan \frac{\pi}{4} - \tan 0\right)$$

$$= \sqrt{2} - 1 - 1 - 0$$

$$= \sqrt{2} - 2$$

Q.27 $\int_0^{\frac{\pi}{4}} \frac{1}{1+\sin x} dx$

Solution:

$$\int_0^{\frac{\pi}{4}} \frac{1}{1+\sin x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{1+\sin x} \times \frac{1-\sin x}{1-\sin x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{1-\sin x}{1-\sin^2 x} dx = \int_0^{\frac{\pi}{4}} \frac{1}{\cos^2 x} dx - \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos^2 x} dx$$

$$= \int_0^{\frac{\pi}{4}} \sec^2 x dx - \int_0^{\frac{\pi}{4}} \sec x \tan x dx$$

$$= [\tan x]_0^{\frac{\pi}{4}} - [\sec x]_0^{\frac{\pi}{4}}$$

$$= \tan \frac{\pi}{4} - \tan 0 - \left(\sec \frac{\pi}{4} - \sec 0 \right)$$

$$= 1 - 0 - \sqrt{2} + 1 = 2 - \sqrt{2}$$

Q.28 $\int_0^1 \frac{3x}{\sqrt{4-3x}} dx$

Solution:

$$\int_0^1 \frac{3x}{\sqrt{4-3x}} dx$$

$$= -\int_0^1 \frac{-3x}{\sqrt{4-3x}} dx$$

$$= -\int_0^1 \frac{4-3x-4}{\sqrt{4-3x}} dx$$

$$= -\int_0^1 \left(\frac{4-3x}{\sqrt{4-3x}} - \frac{4}{\sqrt{4-3x}} \right) dx$$

$$= -\int_0^1 \sqrt{4-3x} dx + 4 \int_0^1 \frac{1}{\sqrt{4-3x}} dx$$

$$= -\int_0^1 (4-3x)^{\frac{1}{2}} dx + 4 \int_0^1 (4-3x)^{-\frac{1}{2}} dx$$

$$= \frac{1}{3} \int_0^1 (4-3x)^{\frac{1}{2}} (-3) dx - \frac{4}{3} \int_0^1 (4-3x)^{-\frac{1}{2}} (-3) dx$$

$$= \frac{1}{3} \left[\frac{(4-3x)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} \right]_0^1 - \frac{4}{3} \left[\frac{(4-3x)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} \right]_0^1$$

$$= \frac{2}{9} \left[(4-3)^{\frac{3}{2}} - (4-0)^{\frac{3}{2}} \right] - \frac{8}{3} \left[(4-3)^{\frac{1}{2}} - (4-0)^{\frac{1}{2}} \right]$$

$$= \frac{2}{9} \left[(1)^{\frac{3}{2}} - (2^2)^{\frac{3}{2}} \right] - \frac{8}{3} \left[(1)^{\frac{1}{2}} - (2^2)^{\frac{1}{2}} \right]$$

$$= \frac{2}{9} [1-8] - \frac{8}{3} [1-2]$$

$$= \frac{2}{9} (-7) - \frac{8}{3} (-1)$$

$$= \frac{-14}{9} + \frac{8}{3} = \frac{-14+24}{9} = \frac{10}{9}$$

Q.29 $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x(2+\sin x)} dx$

Solution:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x(2+\sin x)} dx$$

Put $\sin x = t \Rightarrow \cos x dx = dt$

when $x \rightarrow \frac{\pi}{6}, t \rightarrow \frac{1}{2}$

when $x \rightarrow \frac{\pi}{2}, t \rightarrow 1$

$$= \int_{\frac{1}{2}}^1 \frac{dt}{t(2+t)}$$

Let $\frac{1}{t(2+t)} = \frac{A}{t} + \frac{B}{2+t} \dots$ (i)

Multiplying both sides by $t(2+t)$

$$1 = A(2+t) + B(t) \dots$$
 (ii)

Put $t = 0$ in eq. (ii)

$$1 = A(2) \Rightarrow \boxed{A = \frac{1}{2}}$$

Put $t + 2 = 0 \Rightarrow t = -2$ in eq. (ii)

$$1 = A(0) + B(-2) \Rightarrow \boxed{B = \frac{-1}{2}}$$

Put values of A and B in eq. (i)

$$\frac{1}{t(2+t)} = \frac{1}{2t} - \frac{1}{2(2+t)} \dots$$
 (iii)

$$\int_{\frac{1}{2}}^1 \frac{dt}{t(2+t)}$$

$$= \frac{1}{2} \int_{\frac{1}{2}}^1 \frac{1}{t} dt - \frac{1}{2} \int_{\frac{1}{2}}^1 \frac{1}{2+t} dt \text{ (by (iii))}$$

$$= \frac{1}{2} \left[\ln t \right]_{\frac{1}{2}}^1 - \frac{1}{2} \left[\frac{1}{2} \ln(2+t) \right]_{\frac{1}{2}}^1$$

$$\begin{aligned}
&= \frac{1}{2} \left(\ln 1 - \ln \frac{1}{2} \right) - \frac{1}{2} \left(\ln(2+1) - \ln \left(2 + \frac{1}{2} \right) \right) \\
&= \frac{1}{2} \left(0 - \ln \frac{1}{2} \right) - \frac{1}{2} \left(\ln 3 - \ln \frac{5}{2} \right) \quad \because \ln 1 = 0 \\
&= \frac{-1}{2} \ln \frac{1}{2} - \frac{1}{2} \ln 3 + \frac{1}{2} \ln \frac{5}{2} \\
&= \frac{1}{2} \left(\ln \frac{5}{2} - \ln \frac{1}{2} - \ln 3 \right) \\
&= \frac{1}{2} \ln \left[\frac{\left(\frac{5}{2} \right)}{\left(\frac{1}{2} \right)(3)} \right] \\
&= \frac{1}{2} \ln \left(\frac{5}{3} \right)
\end{aligned}$$

Q.30 $\int_0^{\frac{\pi}{2}} \frac{\sin x}{(1+\cos x)(2+\cos x)} dx$

Solution:

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{(1+\cos x)(2+\cos x)} dx$$

Put $\cos x = t \Rightarrow -\sin x dx = dt$
 $\Rightarrow \sin x dx = -dt$

when $x \rightarrow 0$, $t \rightarrow 1$

when $x \rightarrow \frac{\pi}{2}$, $t \rightarrow 0$

$$= \int_1^0 \frac{-dt}{(1+t)(2+t)}$$

$$= \int_0^1 \frac{dt}{(1+t)(2+t)}$$

Let

$$\frac{1}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t} \dots (i)$$

Multiplying both side by $(1+t)(2+t)$

$$1 = A(2+t) + B(1+t) \dots (ii)$$

Put $1+t=0 \Rightarrow t=-1$ in eq. (ii)

$$1 = A(2-1) + B(0)$$

$$\boxed{A=1}$$

Put $2+t=0 \Rightarrow t=-2$ in eq. (ii)

$$1 = A(2-2) + B(1-2)$$

$$1 = B(-1)$$

$$\boxed{B=-1}$$

Put values of A and B in eq. (i)

$$\frac{1}{(1+t)(2+t)} = \frac{1}{1+t} - \frac{1}{2+t} \dots (iii)$$

Consider

$$\int_0^1 \frac{1}{(1+t)(2+t)} dt$$

$$= \int_0^1 \frac{1}{1+t} dt - \int_0^1 \frac{1}{2+t} dt \quad (\text{by (iii)})$$

$$= [\ln(1+t)]_0^1 - [\ln(2+t)]_0^1$$

$$= \ln(1+1) - \ln(1+0) - [\ln(2+1) - \ln(2)]$$

$$= \ln 2 - \ln 1 - \ln 3 + \ln 2$$

$$= \ln \left(\frac{2 \times 2}{3} \right)$$

$$= \ln \frac{4}{3}$$